Dazaet Galicia-Badillo

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10 October 2026

Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399

Dear Dr Lee,

I am applying for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) on the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 with Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours. My thesis used solid-state NMR with diffraction to relate molecular motion in organic and metal-organic crystals to their macroscopic behaviour.

The chapter closest to this project concerns a solid solution. I synthesised UiO-68-type zirconium MOFs containing a terphenyl rotor linker and a longer azobenzene photoswitch linker in nominal ratios from 10 to 70%. Because the linkers differ in length, the PXRD reflections, which are dominated by the Zr_6 clusters, shifted to lower angle as the photoswitch content increased. I built periodic models with the linkers in ordered positions, and the calculated patterns of the DFT-optimised models contained extra reflections that were absent experimentally, which pointed to a random distribution of linkers within one phase. ^{13}C CP-MAS confirmed that both linkers were incorporated. With a deuterated rotor linker I recorded ^{2}H spectra between 178 and 338 K, which were fitted with 90° and 180° jumps above 10^{7} Hz using electric field gradients calculated in CASTEP. After loading hexafluorobenzene, which I confirmed by ^{19}F MAS and quantified by ^{19}F NMR with standard additions, the 180° jumps were suppressed and the 90° site populations depended on the photoswitch state (ACS Mater. Lett., 2026). In molecular solid solutions of similar-sized components, diffraction changes little with composition, and the local environment seen by solid-state NMR becomes the main source of information.

In a toluene solvate of a pyrrolo[3,2-b]pyrrole that jumps when cooled to 77 K (Mater. Chem. Front., 2026), I prepared the toluene-d_8 analogue and simulated in NMR WebLab the ^{2}H lineshapes recorded from 178 to 338 K, using quadrupolar parameters from GIPAW calculations, which showed methyl rotation and 180° ring flips above 10^{7} Hz. Variable-temperature SCXRD gave a thermal expansion coefficient of 271 MK^{-1} oblique to the face in contact with the cold surface, and we attributed the jump to this anisotropic expansion (with Dr Martin Dračínský and Dr Jan Blahut, IOCB Prague). I was responsible for the NMR analysis, in solution and in the solid state, in all 14 of my publications. For example, I interpreted and modelled the ^{2}H spectra of acetone and tetrahydrofuran in mixed-solvate cocrystals (Angew. Chem. Int. Ed., 2026) and of a guest-loaded zirconium MOF (Chem. Sci., 2020), and interpreted the ^{13}C CP-MAS data of a thermosalient crystal (Chem. Mater., 2025).

I ran my own experiments on a Bruker 500 MHz spectrometer, from acquisition to interpretation and modelling, and also acquired ^{31}P and ^{15}N CP-MAS spectra and analysed ^{13}C T_1 and solid-state ^{1}H and ^{19}F relaxometry data. I made the compounds in my thesis, including the deuterated analogues, and characterised them by SCXRD, PXRD, DSC and TGA. I have not yet used DNP. The UK DNP MAS NMR Facility in Manchester, and the chance to extend this work to ^{17}O and natural-abundance ^{15}N, are among my main reasons for applying.

I co-supervised four bachelor's theses and nine research placements and managed purchasing and inventory for our laboratory. I currently work at the Mexican Institute of Industrial Property until December 2026 and could start in January 2027. Thank you for considering my application.

Yours sincerely,

Dazaet Galicia-Badillo, PhD



Dazaet Galicia-Badillo

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About me

My research has been focused on the design and synthesis of organic crystals and metal-organic frameworks exhibiting dynamics from the molecular level to the macroscopic level, including rotational motion, photoisomerization, crystal flexibility, salient mechanical responses, and single-crystal-to-single-crystal phase transitions. I bring a strong experimental toolkit: organic synthesis, multinuclear solid-state NMR (^{13}C, ^{19}F, ^{31}P, ^{2}H, ^{15}N), single-crystal and powder X-ray diffraction (SCXRD/PXRD), fluorimetry, and DFT calculations. My long-term goal is to build an independent research program that bridges materials chemistry with applied challenges in energy storage, environmental remediation, or biomedical applications in academia or the private sector.

Honors, Awards & Funding

Nominated* Alfonso Caso Medal nomination - UNAM. 2026
Nominated for recognition of outstanding postgraduate achievement (*outcome pending).

Ph.D. awarded with honors Chemical Sciences - UNAM.
2026

Best Presentation Award - IX Seminario Regional de Materiales Avanzados, UAEH, Mexico, October
2025

Supercomputing resource allocations – National High-Performance Computing Laboratory Search (LANCAD)
2024
Prepared successful proposals securing two allocations of computational resources for research.

Best Poster Award - DynaMIC23, Fribourg, Switzerland
2023
Dynamic Materials, Crystals and Phenomena Conference hosted by Prof. Simon Krause

Doctoral scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2022
Support for Ph.D. studies

Master's scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2020
Support for Master's studies

Institute of Chemistry Undergraduate Scholarship -UNAM
2019
Awarded for the best undergraduate research project at the Institute of Chemistry

Education

PhD Chemical Sciences - UNAM, Institute of Chemistry 2022 - 2026
Thesis: Molecular and Macroscopic Motion in Organic Crystals and Metal-Organic Frameworks
Advisor: Prof. Dr. Braulio Rodríguez-Molina

M.Sc. in Chemical Sciences - UNAM, Institute of Chemistry 2020 - 2022

B.S. in Chemistry - UNAM, Faculty of Chemistry 2015 - 2020

Technical skills

Experimental: Organic synthesis, including air- and moisture-sensitive reactions (Schlenk techniques); MOF solvothermal synthesis and post-synthetic modification (PSM). Solid-state NMR (Bruker 500 MHz, double-resonance ^{1}H/X MAS probe): ^{13}C CP-MAS, ^{19}F MAS, ^{31}P, variable-temperature ^{2}H and ^{15}N CP-MAS; ^{2}H lineshape simulation; DFT calculation of NMR parameters (CP2K); analysis of ^{13}C T_1 and solid-state ^{1}H and ^{19}F relaxometry data. X-ray diffraction: SCXRD and PXRD, including variable-temperature studies and structure refinement (ShelX). Photophysics: steady-state and time-resolved fluorescence, UV–Vis absorption. Thermal analysis: hot-stage microscopy, DSC, TGA. **Computational & Software:** DFT calculations (geometry optimization, electronic structure, energy profiles, PES) in Gaussian16; periodic DFT in CP2K and molecular dynamics simulations (LAMMPS) on HPC clusters. Crystallography tools: ShelX, CrystalDiffract, CCDC suite. Visualization: CrystalMaker, ChemDraw. NMR processing and simulation: TopSpin, MestreNova, NMR WebLab, Spinach. AI-assisted workflows for research automation and data analysis. **Lab management:**

Purchasing and vendor management; inventory control and lab organization; team leadership and training; funding acquisition (grants/proposals).

PUBLICATIONS (RESEARCH, REVIEW AND PREVIEW)

14. Ernesto Enriquez-Palacios, **Dazaet Galicia-Badillo**, Marcos Flores-Alamo, Jose Luis Belmonte-Vazquez. Imidazole-based Photoswitches by Design and their use for Data Encryption. *ACS Applied Materials & Interfaces*. **2026.** 18, 35, 47895–47906. *[IF: 8.5]*

13. Dazaet Galicia-Badillo, Ernesto A. Hernández-Morales, Yesterday López-Herrera, Erick Hernández-Santiago, Jan Blahut, Rubén A. Toscano, Martin Dracínský, Braulio Rodríguez-Molina. Dissecting Molecular and Macroscopic Dynamics in a Jumping Solvate. *Materials Chemistry Frontiers,* **2026**. 10, 18, 3305-3313. *[IF: 8.1]*

12. Dazaet Galicia-Badillo, Jade Cisneros, Jan Blahut, Alonso Acosta-Vera, Alfonso García-Márquez, Haggeo Desirena, Erika Bartůnková, Ma. Carmen García-González, Ingrid Chávez-Serrano, Joelis Rodríguez-Hernández, Martin Dracínský, Braulio Rodríguez-Molina. Integration of a Molecular Rotor and a Photoswitch in a Metal-Organic Framework. *ACS Materials Letters*, **2026**. 8, 7, 1995-2001. *[IF: 8.7]*

11. Ernesto A. Hernández-Morales, Dayra Barreto-Hernández, **Dazaet Galicia-Badillo**, Rubén A. Toscano, M. Elena García-Aguilera, Braulio Rodríguez-Molina. Crystals as Rockets: Modulation of the Salient Temperature in Cocrystals by Solvent Mixture Composition. *Angewandte Chemie Int. Ed.*, **2026**, e26010. *[IF: 16.9]*

10. Ángel D. Álvarez-Castillo, Alexis Juárez-Morales, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, José M. Heredia-Peñaloza, Marcos Flores-Álamo, Jose Luis Belmonte-Vázquez. Rational Modulation of Excited-State Intramolecular Proton Transfer, Aggregation-Induced Emission, and Intramolecular Motion on Polysubstituted Imidazoles (Front Cover). *ChemPhotoChem*, **2026**, 10, e70099. *[IF: 3.0]*

9. Mario Castañón-García, Pedro López-Mendoza, **Dazaet Galicia-Badillo**, Luis D. Miranda. Synthesis of Non-Symmetrical 3,3'-Bicoumarins: Total Synthesis of Arteminorin C, 3,3'-Biisofraxidin and Biscopoletin. *Organic Letters*, **2025**, 27 (21), 5349-5354. *[IF: 5.0]*

8. José L. Belmonte-Vázquez, Durga Prasad Karothu, Carl H. Fleischer, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, Robert W. Schurko, Liang Li, Pance Naumov, Braulio Rodríguez-Molina. Low-Temperature Dynamics at Nano- and Macroscales: Organic Crystal That Exhibits Low-Temperature Molecular Motion and the Thermosalient Effect. *Chemistry of Materials*, **2025**, 37, 3373-3383. *[IF: 7.2]*

7. Dazaet Galicia-Badillo, Braulio Rodríguez-Molina. On the Flexibility of 1,4-Bis(phenylethynyl)benzene Butyl Ester Organic Crystal Featuring Mobile Components. *Tetrahedron*, **2024**, 167, 134245. *[IF: 2.2]*

6. Dazaet Galicia-Badillo, José L. Belmonte-Vázquez, Mario Rodríguez, Braulio Rodríguez-Molina, Ma. Carmen García-González. Aggregation-Induced Emission Enhancement and Solid-State Photoswitching of Crystalline Carbazole N-Salicylidene Anilines. *ACS Omega*, **2024**, 9 (36), 38015-38022. *[IF: 4.3]*

5. Alonso Acosta-Vera, **Dazaet Galicia-Badillo**, Armando Navarro-Huerta, Konstantin Stracke, Virginia Gómez-Vidales, Joelis Rodríguez-Hernández, Jack D. Evans, Braulio Rodríguez-Molina. Modulation of the Photophysics and Internal Dynamics in a Zr Metal-Organic Framework by the Inclusion of Fluorinated Guests. *ACS Materials Letters*, **2024**, 6 (9), 4395-4401. *[IF: 8.7]*

4. Dazaet Galicia-Badillo, Paulina Noemí García-Vargas, Braulio Rodríguez-Molina. Preview: The Integration of Multiple Components for Multi-Responsive Dynamic Framework Materials. *Matter*, **2024**, 7 (8), 2766-2768. *[IF: 17.5]*

3. Erick Hernández-Santiago, Jorge Espinosa-Rocha, **Dazaet Galicia-Badillo**, Braulio Rodríguez-Molina. Review: Local and Global Dynamics in Organic Soft Porous Crystals. *Matter*, **2024**, 7 (2), 408-429. *[IF: 17.5]*

2. Manuel Rodríguez-Molina, **Dazaet Galicia-Badillo**, Enoc Cetina-Mancilla, Jorge Cárdenas, Lilian I. Olvera, Rubén A. Toscano, Braulio Rodríguez-Molina, Mikhail G. Zolotukhin. 9-Trifluoromethylxanthenediols: Synthesis and Supramolecular Motifs. *ACS Omega*, **2022**, 7 (16), 13520-13528. *[IF: 4.1]*

1. Aaron Torres-Huerta, **Dazaet Galicia-Badillo**, Andrés Aguilar-Granda, Jacob T. Bryant, Fernando J. Uribe-Romo, Braulio Rodríguez-Molina. Multiple Rotational Rates in a Guest-Loaded, Amphidynamic Zirconia Metal-Organic Framework. *Chemical Science*, **2020**, 11 (42), 11579-11583. *[IF: 9.8]*

TEACHING & MENTORSHIP

Co-director of 4 bachelor's theses (2023-present), covering MOF synthesis, organic crystal dynamics, and luminescent materials. Supervisor of 9 undergraduate short research stays (2022-2025) and 1 high school student.

Scientific conferences & presentations

15 posters (6 international, 9 national) · 9 oral presentations (4 invited lectures, 5 congress talks)

Scientific divulgation

1 article (Boletín IQ-UNAM, 2022) · 2 public lectures (High Schools, 2024–2025) · 1 media consultation (UNAM Global, 2022)

References

Prof. Dr. Braulio Rodríguez Molina - PhD Supervisor**.** Instituto de Química, Universidad Nacional Autónoma de México
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